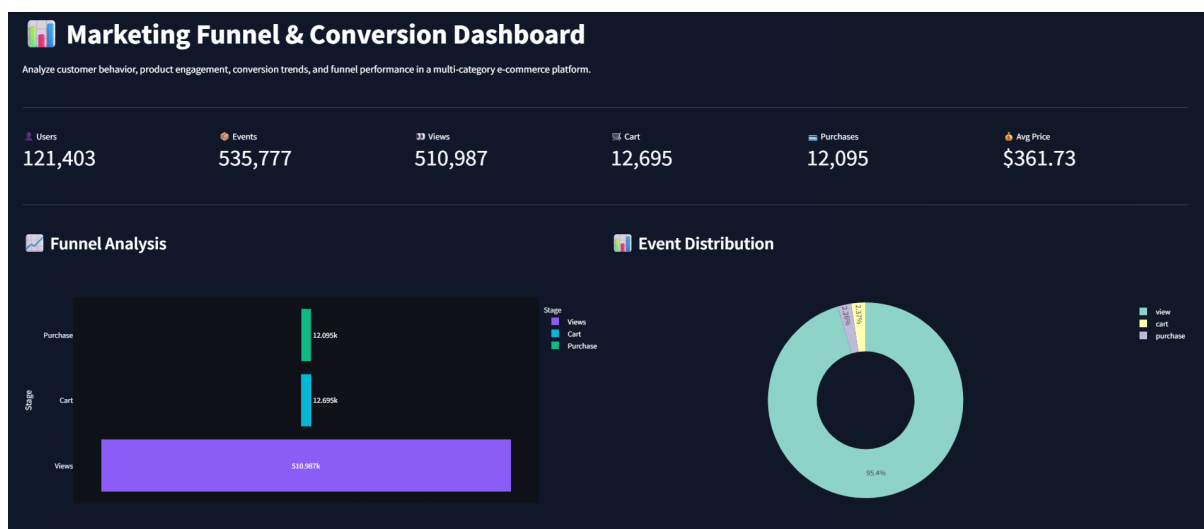


Marketing Funnel & Conversion Performance Analysis

Future Interns - Data Science & Analytics Internship

Task 3 Report



Submitted By:

Purvi Jain

CIN ID: FIT/APR26/DS16984

Tools Used:

Python, Pandas, Streamlit, Plotly, Seaborn

1. Abstract

This project focuses on analyzing customer behavior and marketing funnel performance using Python-based exploratory data analysis and an interactive Streamlit dashboard. The analysis was performed on a large-scale multi-category e-commerce dataset containing customer interaction events such as product views, cart additions, and purchases.

The project aims to identify customer conversion trends, funnel drop-offs, product engagement patterns, and purchase behavior. Multiple interactive visualizations were developed to transform raw transactional data into actionable business insights.

The dashboard provides an overview of key business metrics including total users, views, cart events, purchases, category performance, and conversion analysis.

2. Introduction

Marketing funnel analysis plays a significant role in understanding customer journeys and improving business conversion strategies. In the e-commerce industry, analyzing customer interactions helps businesses optimize product recommendations, improve checkout processes, and increase overall sales performance.

This project presents an interactive Python dashboard developed using Streamlit and Plotly for analyzing customer behavior and funnel performance.

The dashboard highlights:

- Funnel conversion analysis
- Event distribution analysis
- Product category performance
- Brand performance analysis
- User activity trends
- Purchase behavior analysis
- Daily customer activity insights

The dashboard enables businesses to monitor customer engagement and identify areas for improving conversion rates and reducing customer drop-offs.

3. Objectives

The main objectives of this project are:

1. To analyze customer interaction behavior on the e-commerce platform.
2. To identify conversion drop-offs in the marketing funnel.
3. To study product category and brand performance.
4. To analyze purchase trends and customer engagement.
5. To generate business insights using exploratory data analysis.
6. To create an interactive and visually appealing dashboard using Python.

4. Tools and Technologies Used

Tool	Purpose
Python	Data analysis and dashboard development
Pandas	Data cleaning and preprocessing
Streamlit	Interactive dashboard creation
Plotly	Interactive visualizations
Seaborn	Statistical data visualization
NumPy	Numerical operations

5. Dataset Description

The dataset used in this project is the **E-Commerce Behavior Dataset from Multi-Category Store**.

Dataset Source:

<https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store>

The dataset contains customer interaction records including:

- Event timestamps
- Product categories
- Brand information
- Product prices
- User IDs
- Event types:
 - View
 - Cart

– Purchase

The dataset contains millions of customer interaction records collected from an e-commerce platform.

6. Data Cleaning Process

The following preprocessing steps were performed:

1. Removed duplicate records.
2. Handled missing values in brand and category columns.
3. Converted event timestamps into datetime format.
4. Extracted hour, date, and weekday information.
5. Processed event types for funnel analysis.
6. Prepared aggregated datasets for dashboard visualizations.

7. Dashboard Overview

The dashboard was developed using Streamlit with a modern dark-themed user interface.

The dashboard contains multiple visualizations to analyze customer behavior and marketing funnel performance effectively.

The dashboard includes:

- KPI Cards
- Funnel Analysis Chart
- Event Distribution Pie Chart
- Price Distribution Histogram
- Top Categories Analysis
- Brand Performance Treemap
- Purchases by Hour Line Chart
- Category vs Event Type Analysis
- User Activity Scatter Plot
- Daily Active Users Area Chart

- Average Price Analysis

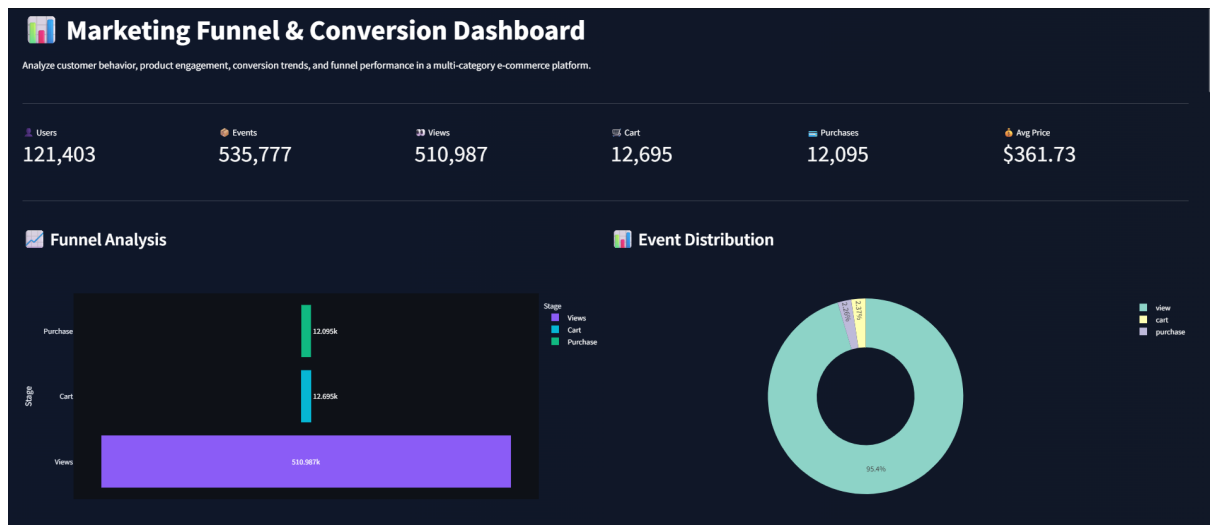


Figure 1: Marketing Funnel & Conversion Dashboard

8. KPI Analysis

The dashboard highlights the following Key Performance Indicators (KPIs):

KPI	Value
Total Users	121,403
Total Events	535,777
Total Views	510,987
Total Cart Events	12,695
Total Purchases	12,095
Average Product Price	\$361.73

9. Visualization Analysis

9.1 Funnel Analysis

The funnel visualization shows customer movement across the following stages:

- Product View
- Add to Cart
- Purchase

A significant customer drop-off was observed between product viewing and purchase stages.

9.2 Event Distribution

The event distribution chart indicates that product views dominate customer interactions, while purchases represent a smaller portion of total events.

9.3 Product Category Analysis

Certain product categories generated significantly higher engagement and purchase activity compared to others.

9.4 Brand Performance Analysis

The treemap visualization highlights the most interacted and purchased brands on the platform.

9.5 Purchases by Hour

The hourly purchase trend analysis shows peak customer purchasing activity during specific hours of the day.

9.6 User Activity Analysis

The user activity analysis identified highly active users and daily customer engagement trends across the platform.

10. Key Insights

- Product views are significantly higher than completed purchases.
- Major customer drop-off occurs between view and cart stages.
- Cart abandonment affects conversion performance.
- Mid-range priced products receive better customer engagement.
- Certain product categories dominate overall platform activity.
- Peak purchase activity occurs during specific hours.
- Some brands achieve consistently higher interaction rates.

11. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Improve product page design and product descriptions.
2. Simplify the checkout process to reduce cart abandonment.
3. Use retargeting campaigns for abandoned cart users.
4. Focus marketing strategies on high-performing categories.
5. Run promotional campaigns during peak activity hours.
6. Implement personalized product recommendation systems.
7. Analyze customer behavior further for customer segmentation.

12. Challenges Faced

- Handling large-scale datasets with millions of records.
- Managing missing values and inconsistent category data.
- Optimizing dashboard performance for faster rendering.
- Designing visually balanced and interactive layouts.
- Creating meaningful funnel conversion metrics.

13. Learnings

Through this project, the following skills were developed:

- Exploratory Data Analysis (EDA)
- Customer Funnel Analysis
- Interactive Dashboard Development
- Data Cleaning and Preprocessing
- Business Insight Generation
- Python Data Visualization
- Streamlit Application Development
- Plotly Interactive Charting

14. Conclusion

This project successfully analyzed customer behavior and marketing funnel performance using Python and interactive dashboarding techniques.

The dashboard provides valuable insights into customer engagement, purchase behavior, conversion trends, and product performance.

The analysis demonstrates the importance of data-driven decision-making in improving customer experience, reducing funnel drop-offs, and increasing business conversions.

The project also highlights the practical applications of Python-based analytics and dashboard development in solving real-world business problems.

15. References

1. Python Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. Plotly Documentation
5. Kaggle E-Commerce Dataset
6. Future Interns Data Science and Analytics Internship Guidelines